



6.3 Winder Suppliers

Winding Process, Roll Structure, Winder Design

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Winder or Slitter-Rewinder Suppliers

- ▶ Accraply (Stanford) – www.accraply.com
- ▶ Ashe Converting (UK) - www.ashe.co.uk
- ▶ Atlas Converting (including Titan) – www.Atlasconverting.com
- ▶ Beloit (closed) – www.paperindustryweb.com (History of Beloit)
- ▶ Birch Brothers - www.BirchBrothers.com
- ▶ Brampton (Canada) - www.BE-ca.com
- ▶ Bruckner (Germany) – www.Bruckner.com
- ▶ Catbridge Machinery – www.Catbridge.com
- ▶ CTC International – www.CTCint.com
- ▶ Davis-Standard (formerly Black-Clawson) – www.Davis-Standard.com
- ▶ Deacro – www.Deacro.com
- ▶ Dimatra Converting (Mexico) – www.Dimatraconverting.com
- ▶ Elite Cameron – www.elitecameron.com
- ▶ Euromac (Italy) - www.Euromacslitters.com
- ▶ Faustel – www.Faustel.com
- ▶ GL&V - www.GLVpulppaper.com
- ▶ Ghezzi E Annoni (Italy) - www.GAweb.net
- ▶ Innovative Machine Corp - www.innovativemach.com
- ▶ Jennerjahn – www.Jennerjahn.com
- ▶ Kampf-Group Jagenberg (Germany) – www.Kampf.de
- ▶ KTI (Keene Technology) – www.keenetech.com
- ▶ Macro Engrg. – www.macroeng.com
- ▶ MTorres (Spain) – www.mtorres.es
- ▶ Monomatic (France) – www.Monomatic.fr
- ▶ New Era Converting – www.NewEraconverting.com
- ▶ Parkinson Technologies - www.Parkinsontechnologies.com
- ▶ Pyradia (Canada) - www.pyradia.com
- ▶ Valmet (Finland) – www.Valmet.com
- ▶ Voith (Germany) – www.Voith.com
- ▶ US Webcon – www.uswebcon.com

- ▶ Valmet Surface Winder Transfer
www.youtube.com/watch?v=LbGMA6rrlfs
- ▶ Voith Two Drum Slitter Rewinder
www.youtube.com/watch?v=Ra87F8Xvy9s
- ▶ Brampton Winder
www.be-ca.com/resources/videos/watch/flexwin-blown-film-winder
- ▶ MTorres Auto-Transfer Surface Winder
www.youtube.com/watch?v=K_p_pfyowAM
- ▶ Jennerjahn Coreless Winder
www.jennerjahn.com/Index/videos.php?v_search=&video=^9s9Hb*G4f
- ▶ Atlas Converting Labelstock Slitter Rewinder
www.youtube.com/watch?v=Cbr115o7U70

Paper Industry Web (PIW)

PAPER INDUSTRY WEB (PIW)

paperindustryweb.com

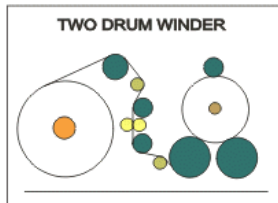
INTRODUCTION TO WINDING

PURPOSE OF THE PAPER MACHINE WINDER

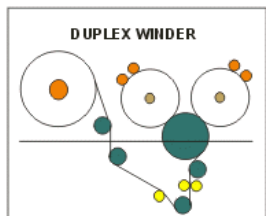
The paper comes off the paper machine in very large rolls called jumbo rolls or reels. The jumbo is the width of the paper machine ranging nominally from 100" to 360" {2540 to 9140 mm} wide. The jumbo roll is wound on a shaft called a reel spool. The reel spool diameter and weight is proportionate to the width of the paper machine and the weight of the jumbo. It is not unusual to see reel spools 30" to 36" diameter {762 to 914 mm} and jumbo reels greater than 100" {2540 mm} diameter. On large paper machines jumbo rolls can weigh in excess of 35 tons {31.7 metric tons}.

The purpose of the machine winder is to convert the large jumbo roll into sizes convenient to ship. The size of the "shipping roll" will be determined by the end use, generally referred to as the converting process. Examples of some of the converting end uses are printing presses, corrugating machines, bag machines, sheeters, etc.

THE TWO BASIC WINDER TYPES.



The most popular type paper machine winder is known as a two drum winder. The two drum winder winds the rolls on a single axis, all the rolls being wound on a common center, adjacent to each other across the width of the winder. Papers wound on a two drum winder are standard newsprint, fine papers such as tablet or copy papers, board grades, tissues, and towel to name a few. Normally, only a single two drum winder is required to handle the paper machine production.

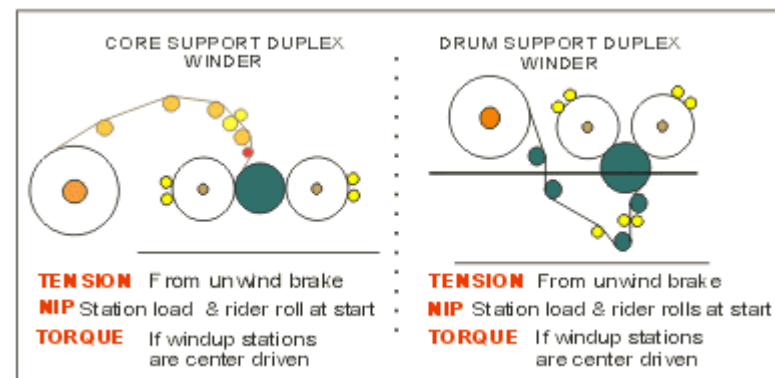
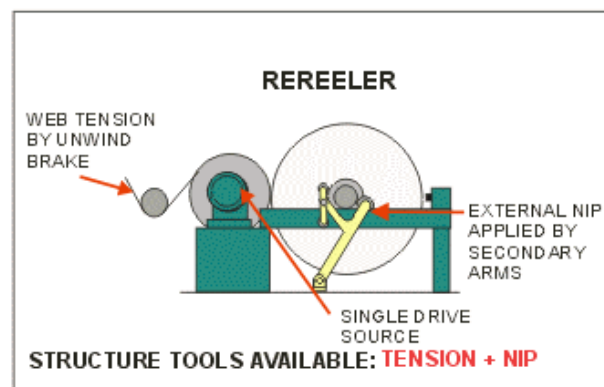
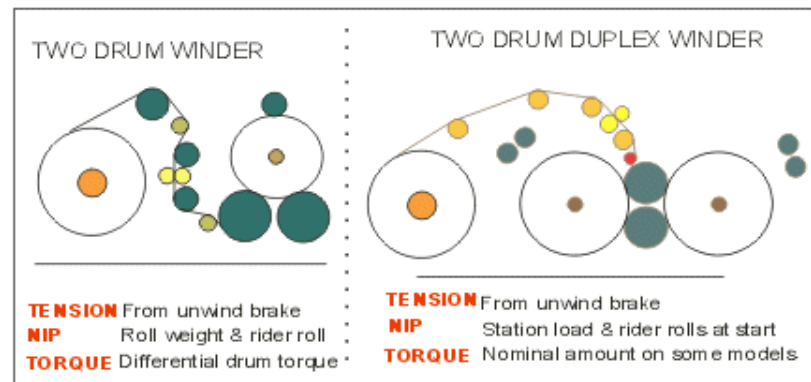
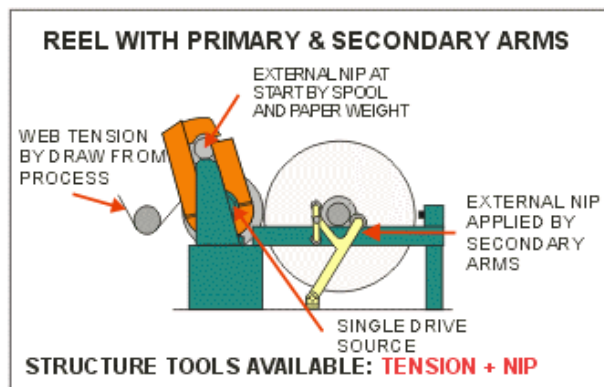


A second popular type paper machine winder is known as a duplex winder, used to wind papers that are considered difficult to wind, that is: To structure the shipping rolls in a manner that the defects in the roll will be reduced. Papers wound on a duplex winder are generally coated papers such as magazine stock, high gloss specialty papers, enhanced newsprint, etc. These are papers that are filled with coating materials and processed in an additional finishing operation to make the surface extremely smooth and glossy. In looking at it will be noted that unlike a two drum winder, each individual roll is supported and wound at separate winding stations located on opposite sides of a center drum across the full winder width.

Paper Industry Web (PIW)



PAPER MACHINE REELS & REREELERS



Accraply (Stanford)



Ashe Converting (UK)



Primary Slitter Rewinders

Width Range:
Max Speed:
Max Diameter:
Material:

- 3.5m to 8.7m
- Up to 1200 m/min
- Up to 1200mm
- Plastic Films



Secondary Slitter Rewinders

- 1.6m to 3.5m
- Up to 1200 m/min
- Up to 1200mm
- Plastic Films/Papers



Labelstock/Paper Slitter Rewinders

- Up to 3m
- Up to 800m/min
- Up to 1800mm
- Labelstock/Papers



Duplex Slitter Rewinders

- 400mm to 2.2m
- Various
- 600mm to 1000mm
- Films/Laminates/Papers/Foils



Duplex Turret Rewinder

- 800mm to 2.2m
- Up to 700m/min
- 450mm and 600mm
- Films/Laminates/Papers/Foils



Rewinding/Doctoring Machines

Width Range:
Max Speed:
Max Diameter:
Material:

- 300mm to 2.5m
- Up to 800 m/min
- Up to 1500mm
- Various



Label Inspection Slitter Rewinders

- Up to 600mm
- Up to 300m/min
- Up to 600mm
- All types of labels



Label Inline and Offline Turret Rewinders

- Up to 600mm
- Up to 250m/min
- Up to 400mm
- Labelstock/Papers



Label Converting Lines

- Up to 600mm
- Up to 250m/min
- Up to 600mm
- Labelstock/Papers



Bespoke

- As required
- As required
- As required
- As required

Atlas Converting



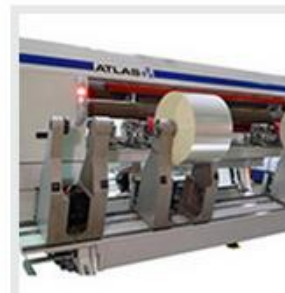
Atlas CW500 >>



Atlas CW800 >>



Atlas CW964 >>



Atlas CW3600 >>



Atlas CW5400 >>



Atlas CW1040 >>



Atlas CS2400 >>



Atlas Uni LS >>



Titan ER610 >>



Titan SR800 >>



Titan SR9-DS >>



Titan SR9-DT >>

Birch Brothers



Brampton Engineering (Canada)



Catbridge



Model 324 Cantilevered Turret Slitter Rewinder



Model 324 FP Single-Turret Slitter Rewinder



Model 344 Dual-Turret Slitter Rewinder



Model 344 FP Dual Turret Slitter Rewinder



Model 334 Fully Automatic Turret Slitter
Rewinder



Model 324-C Continuous Turret Center Rewinder

Catbridge



Model 210 Two-Drum Surface Winder



Model 205 Two-Drum Surface Winder



Model 714 Two-Drum Surface Winder



Model 729 Duplex Center/Surface Slitter
Rewinder



Model 550 Duplex Center/Surface Slitter
Rewinder

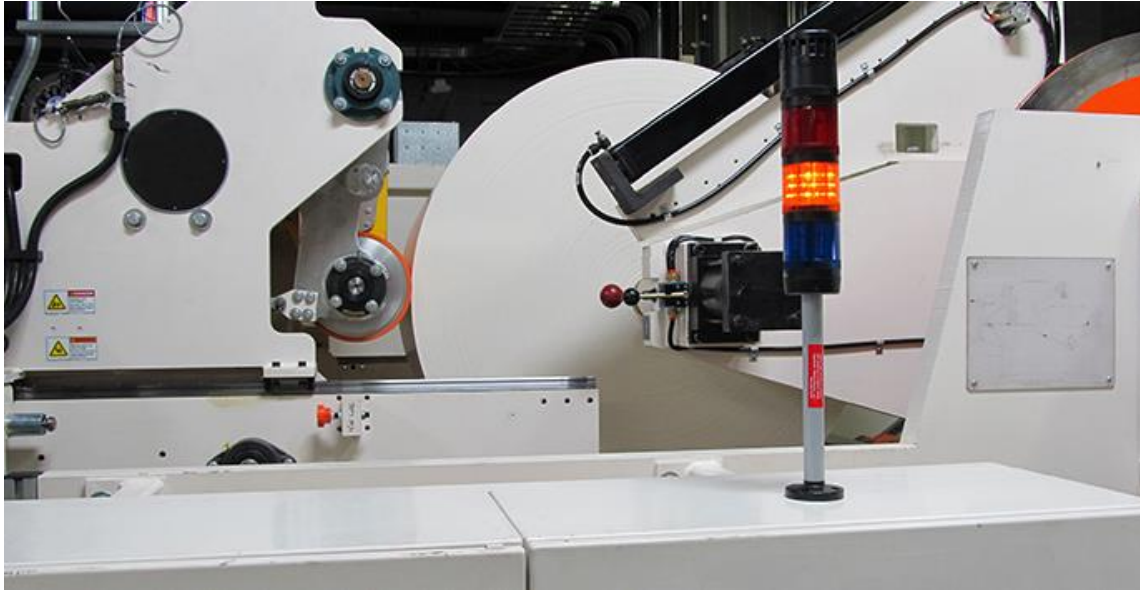


Model 900 Duplex Center Slitter Rewinder

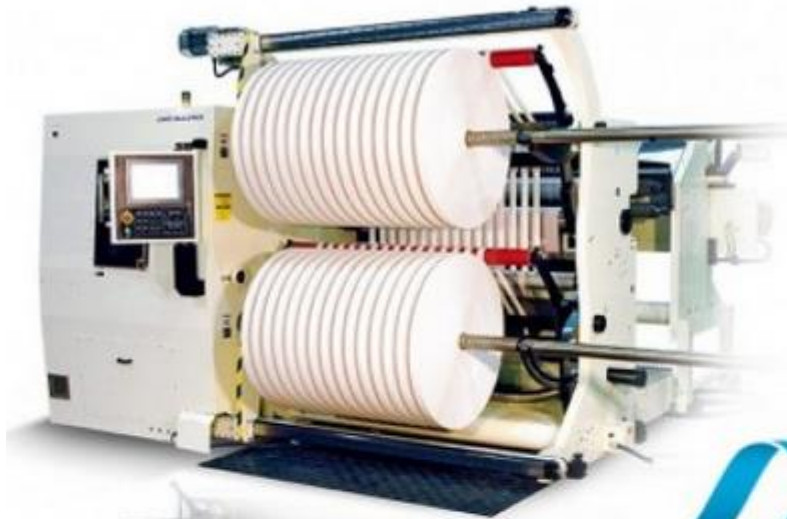
CTC International



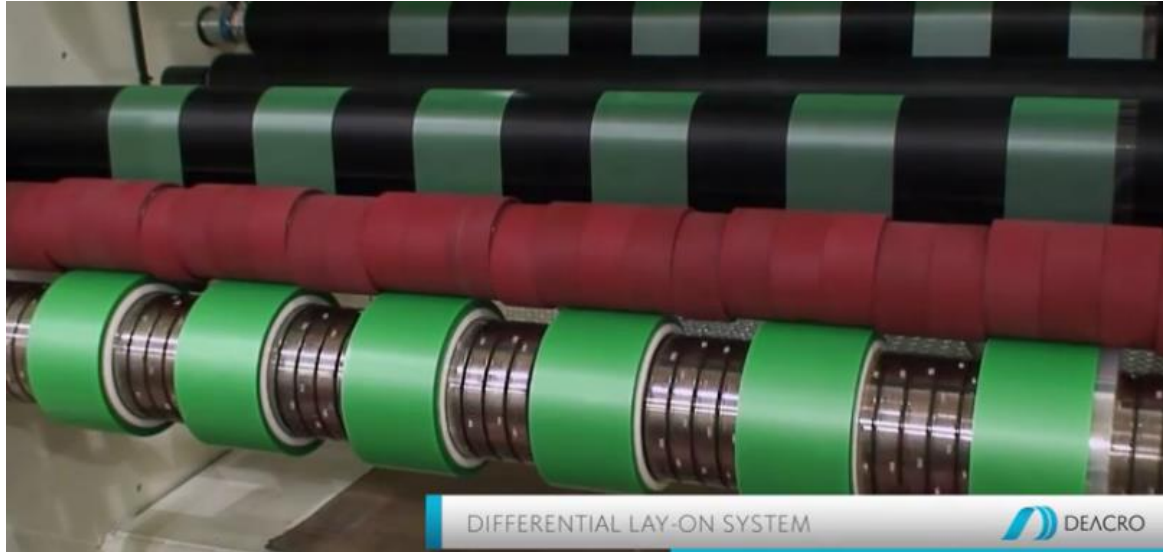
Davis-Standard (formerly Black Clawson)



Deacro (Canada)



Deacro (Canada)



Dimatra Converting (Mexico)



Plastic Film



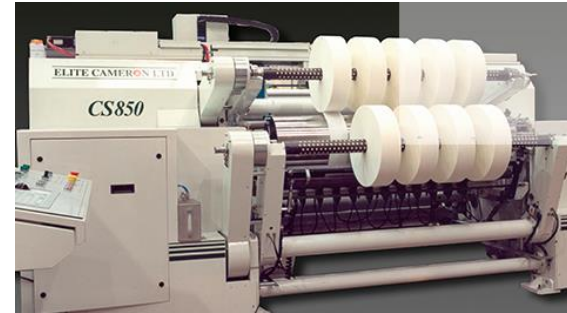
Paper



Nonwoven

Dimatraconverting.com

Elite Cameron



CS7400



TD4300



TD4400



TD4600



CS500



CS600



CS850



CS1000



CW250



CW400



CW600 / CW800

CW600HD /
CW800HD

Euromac (Italy)

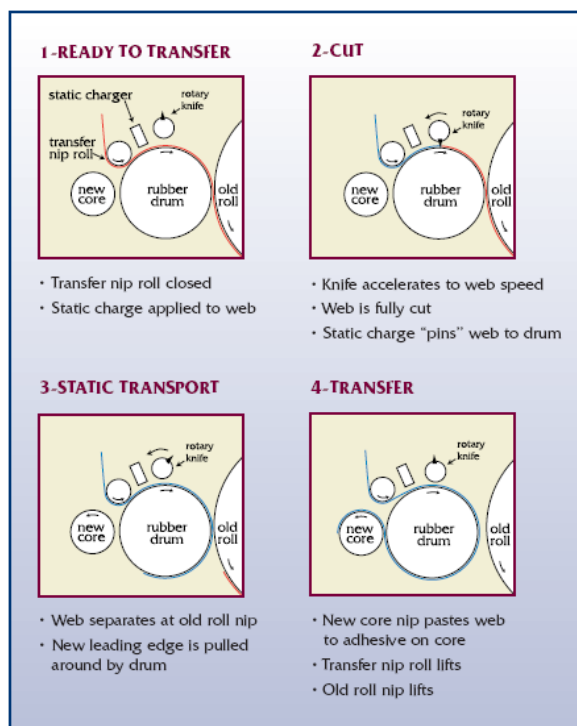


Faustel



Surf-Ctr Winder w/ Auto Transfer

Faustel makes a unique central drum rewriter with at-speed automatic roll transfer from full roll to new core (from spindle A to B back to A, etc.)



Images courtesy of Faustel Corp.

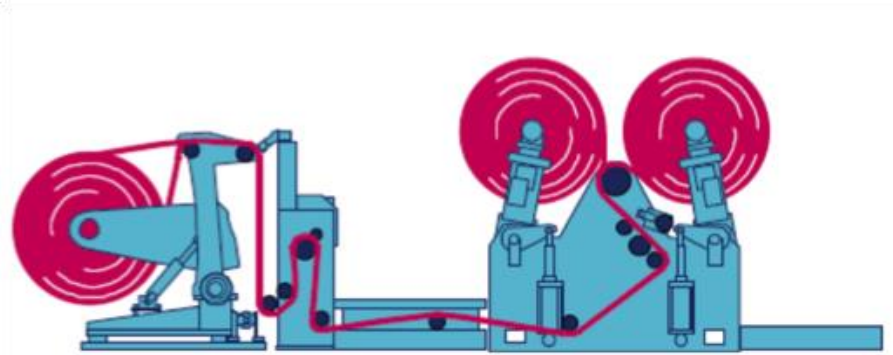
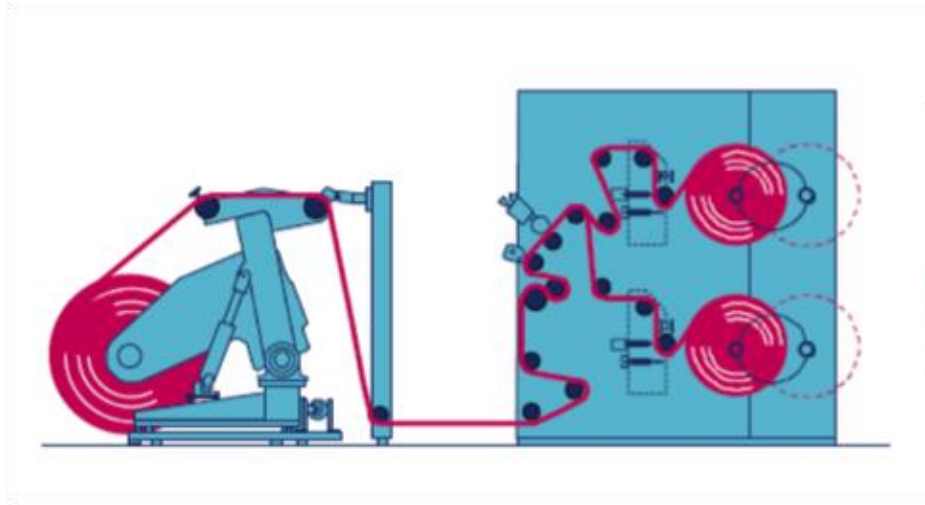
GL&V



Ghezzi E Annoni (Italy)



ilamco (Spain)

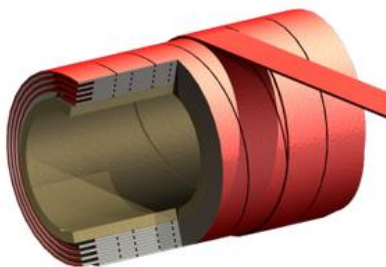


Images courtesy of ilamco - www.ilamco.com

Independent Machine

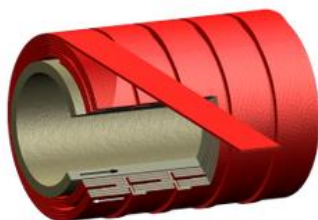
Level Wind

The turnaround ends of each layer are at the same point



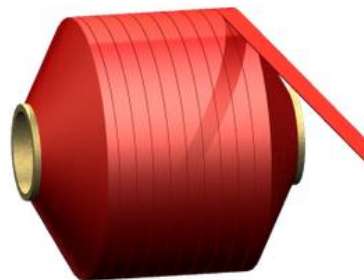
Index Wind

Programmed stacked lanes with precise programmed index to the adjacent lane also called.. step wind

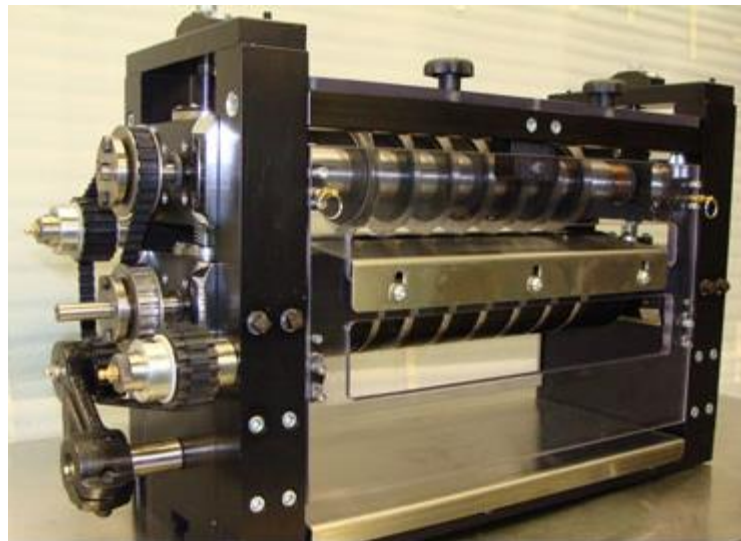
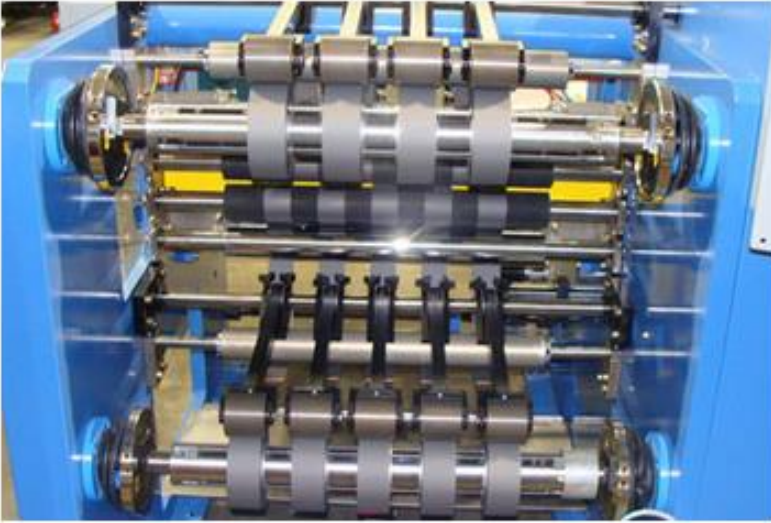


Taper Wind

Each traverse layer tapers a programmed amount from the edge to form a partial pyramidal shape



Innovative Machine Corp



Jennerjahn



Kampf (Jagenberg)

Kampf.de



Autoslit III



BaseLiner



Centurial



CombiLiner



Combiwinder



Conslit



Conslit III



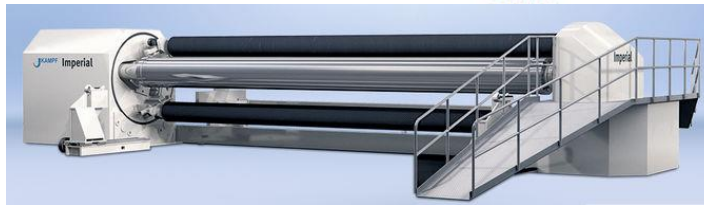
Conslit III
Aluminium



Duplomat



Ecoliner



Inliner



Jumbowinder



KRD



Microslit Capacitor



Microslit Converting



RKW 1003



Roll-O-Fix Evolution



Sepamat



Spulmaschine



Starslit



Trimming Line



Unial / Unial ECO



Unislit II



Unislit II Aluminium



Unislit L



Universal BOPET



Universal BOPP



Universal TF



Battery Applications

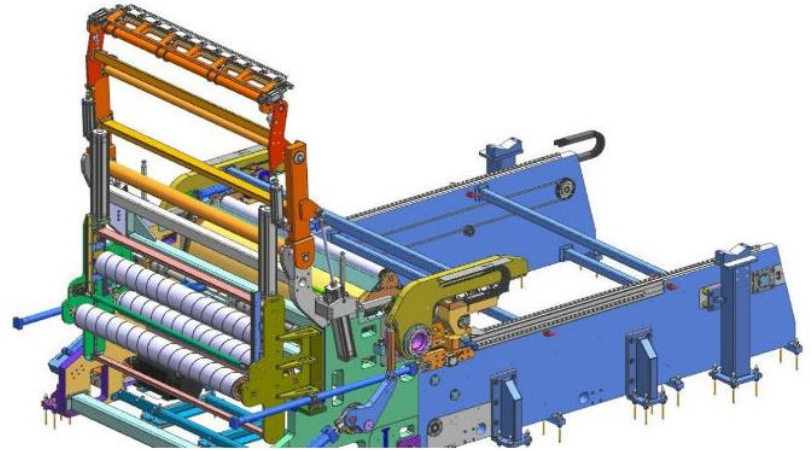
KTI (Keene Tech.)



Macro Engineering



MTorres (Spain)



Monomatic (France)



Single station revider

Full Speed

surface drive

Full Speed

axial drive and surface drive



URS



IPRX



PRC



LR



IMRX



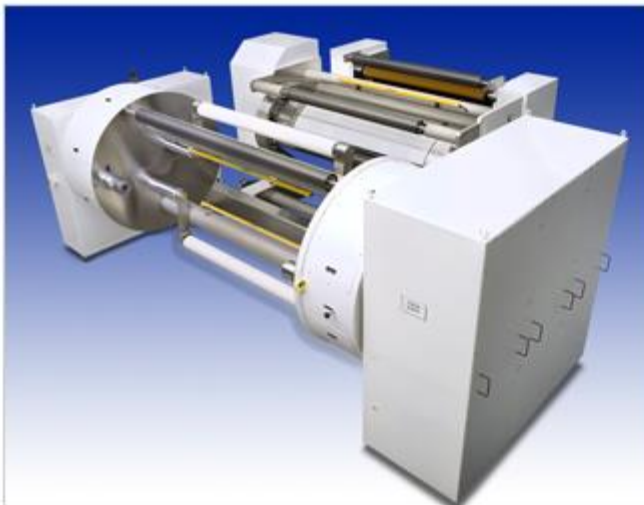
MRC

New Era Machinery



Cleanroom Turret Winders

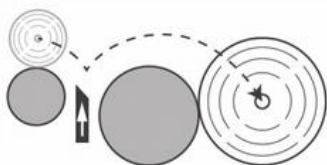
New Era takes great care in the design of its clean room equipment to minimize the number of debris generating components on the equipment, shroud any moving components under enclosures and incorporate special cutting methods designed to reduce debris generated while cutting. All this is accomplished while still allowing the same equipment features as a standard New Era turret winder designed for a general purpose environment.



Zero-Foldover Winders

Our patented zero fold-over winder allows for an automatic on the fly transfer from one roll to the next with the elimination of any tail. This process allows for minimal waste at the core of the roll and utilizes a traversing knife cut-off mechanism to minimize debris generation.

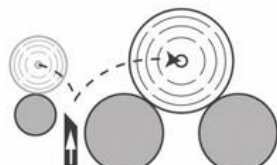
Parkinson Technologies



MkII Horizontal Surface Winder

Automatic surface winding of lofted or nip sensitive materials. Can produce large diameter finished rolls with light wound-in tension to prevent damage to sensitive webs.

- Materials: Nonwovens, Textile, Roofing Membrane, Specialty Film
- Material Gauge: 0.3-7.4 osy (10-250 gsm)
- Web Widths: 20"-157" (0.5-4.0 m)
- Max Roll Diameter: 120" (3.0 m)



MkIII Surface Winder

Automatic surface winding of finished rolls up to 42" diameter. Wound rolls are fully supported, thus inline slitting of wide materials and winding onto small diameter cores is possible.

- Materials: Nonwovens (dense), Textile, Roofing Membrane
- Material Gauge: 0.3-14.7 osy (10-500 gsm)
- Web Widths: 39"-236" (1-6 m)
- Max Roll Diameter: 40"-42" (1 m)



Parkinson Technologies (including Dusenbery)



Advantage Series
Cantilevered Center Slitter Rewinder

- All materials including pressure-sensitive
- Material widths up to 100 inches (2540 mm)
- Speeds up to 2,000 fpm (610 mpm)
- Slitting types: score, shear, razor
- Slit widths from 1 inch (25 mm) to full width
- Rewind diameter 40 inches (1 m)



Genesis Series
Center Driven Duplex Slitter Rewinder

- All materials excluding pressure-sensitive
- Material widths up to 62 inches (1574.8 mm)
- Speeds up to 1,500 fpm (457 mpm)
- Slitting types: score, shear, razor
- Slit widths from .5 inch (12.7 mm) to full width
- Rewind diameter 30 inches (762 mm)



Flex Series
Center Surface Slitter Rewinder

- Label stock, laminates, paperboard, and foams
- Material widths up to 180 inches (4575 mm)
- Speeds up to 2,500 fpm (762 mpm)
- Slitting types: score, shear, razor
- Slit widths from 2 inch (50.8 mm) to full width
- Rewind diameter 50 inches (1270 mm)



Alpha Series
Two Drum Surface Slitter Rewinder

- Large diameter paper and nonwovens
- Material widths up to 160 inches (4 m)
- Speeds to 2,500 fpm (762 mpm)
- Slitting types: shear and score
- Slit widths from 2" (50 mm) to full width
- Maximum rewind diameter 160" (4 m)

Pyradia (Canada)



Automatic surface winder



Single position unwind



Dual position uncoiler



Floor pick-up unwind



Drum surface winder



Open frame turret winder



Phantom axis turret winder



Linear Displacement winder



Unwind cut-to-length



Cantilevered winder



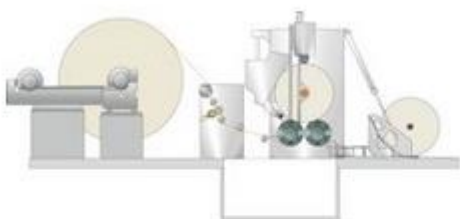
Salvage winder

US Webcon

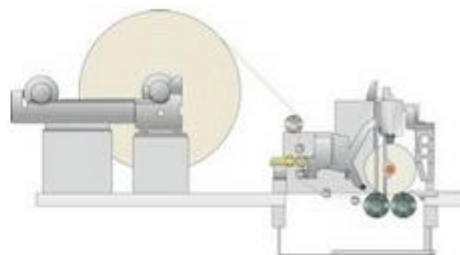


Valmet (Metso)

Two-drum
winders



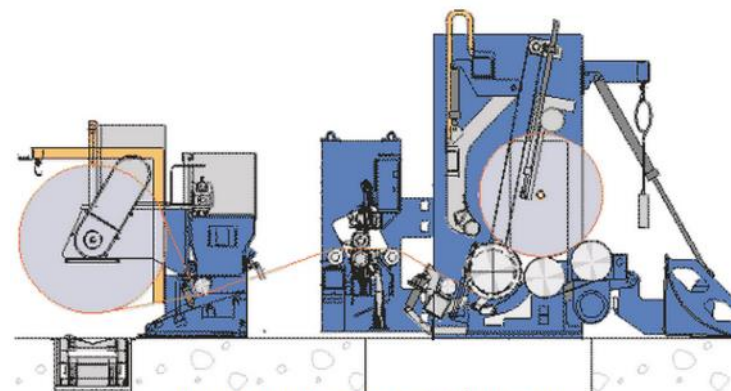
Two-drum base
winders



Pulp winders



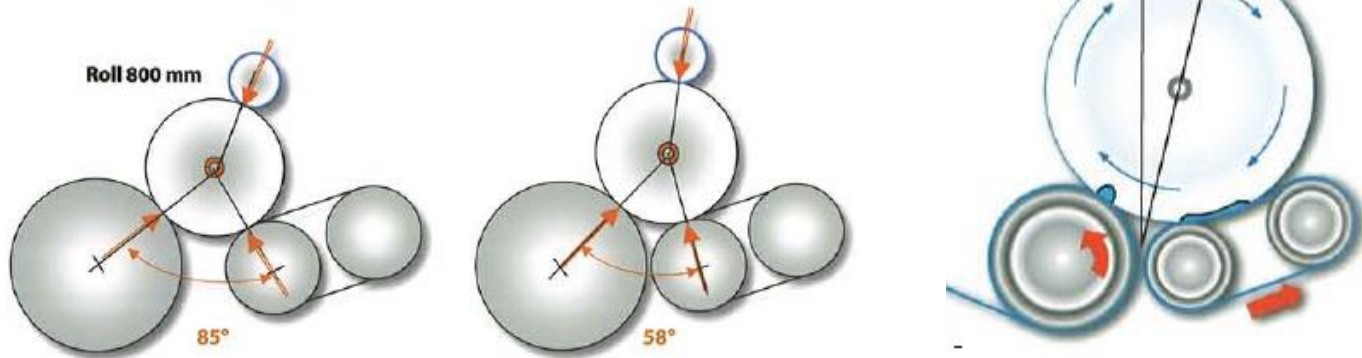
Belt supported
winders



Bobineuse type Winbelt the company Metso Paper

Belt-Drum Surface Rewinder

“Set change in 12 seconds.”



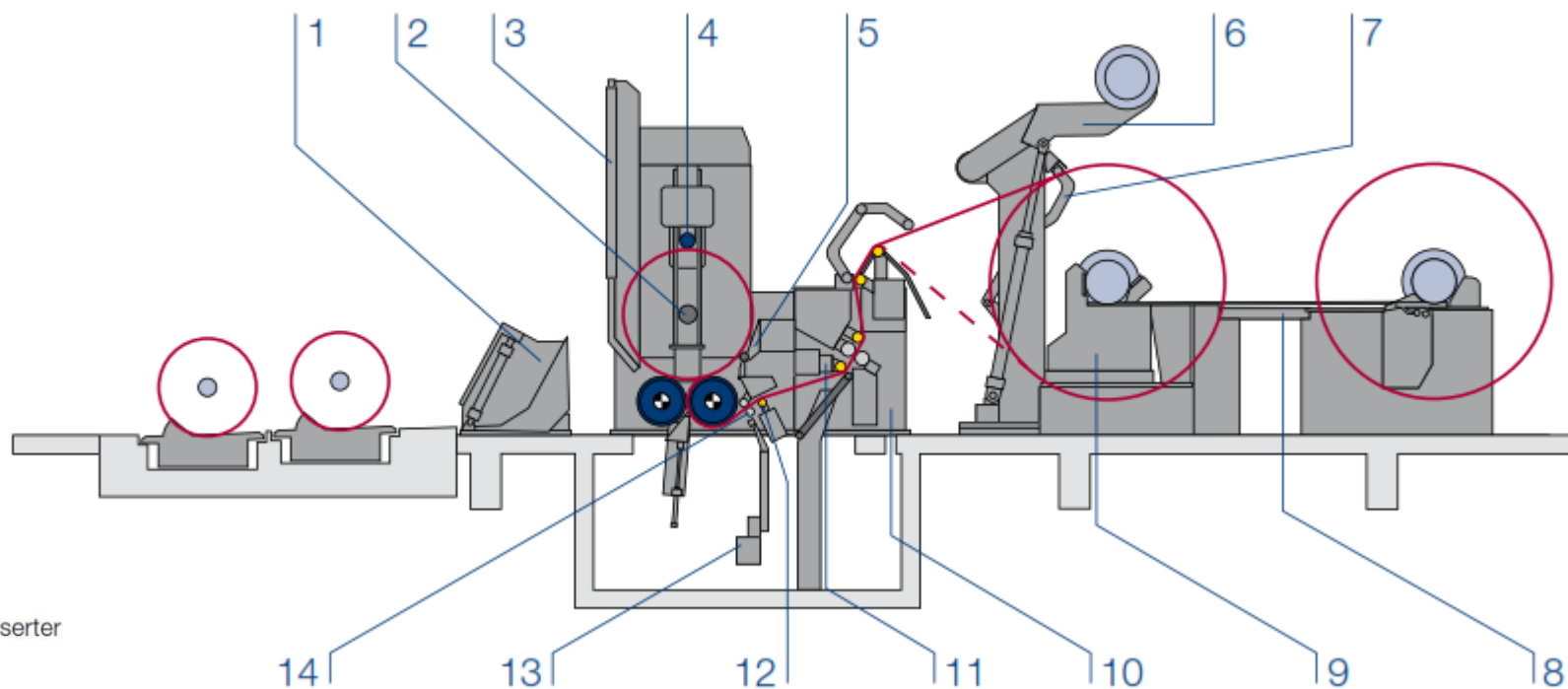
“The WinBelt nip load control prevents roll defects caused by excess nip loads:

Crepe wrinkles on newsprint and catalog papers, corrugation on coated papers, unwanted gloss and mottling on coated boards.”

Images courtesy of Valmet / Metso Paper

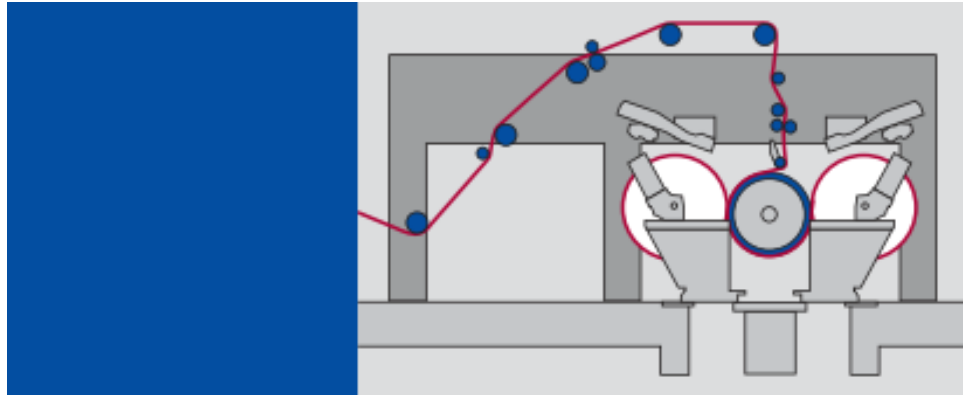
Voith (Germany)

VariFlex Design

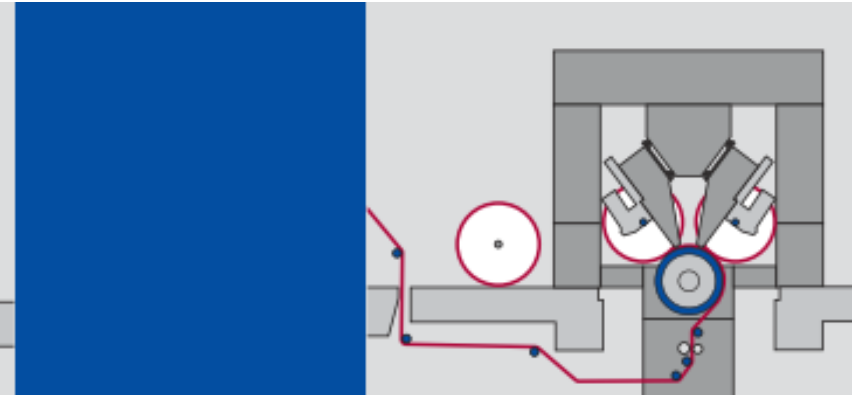


- 1 Lowering platform
- 2 Core chucks
- 3 Safety guard
- 4 Rider roll
- 5 Ejector with core inserter
- 6 Reel spool lift-out
- 7 FlyingSplice
- 8 Parent roll magazine
- 9 Parent roll unwind
- 10 Slitter section
- 11 Web tension measurement
- 12 Spreader unit
- 13 Gecko OneStep (start and end)
- 14 Perforating unit

Voith (Germany)



In the case of the VariPlus™, the paper rolls are wound in the 3 and 9 o'clock position.



In the case of the VariTop™, the paper rolls are wound in the 1 and 11 o'clock position.

